5. (a) $\mathbb{Z}[\sqrt{5}] = \{a + b\sqrt{5} : a, b \in \mathbb{Z}\} \cong \dfrac{\mathbb{Z}[x]}{\langle x^2 - 5\rangle}$

$$\varphi : \mathbb{Z}[x] \longrightarrow \mathbb{Z}[\sqrt{5}]$$
$$f(x) \longmapsto f(\sqrt{5})$$

$$f(x) = g(x)$$
$$\underline{f(\sqrt{5})} = \underline{g(\sqrt{5})}$$?

$$f(x) + g(x) \longmapsto f(\sqrt{5}) + g(\sqrt{5})$$
$$f(x)g(x) \longmapsto f(\sqrt{5})\, g(\sqrt{5})$$

$$a + b\sqrt{5} \longleftarrow\!\shortmid\ \underline{a + bx = f(x)}$$

$$\frac{\mathbb{Z}[x]}{\ker \varphi} \cong \mathbb{Z}[\sqrt{5}] \ [\text{by } 1^{st} \text{ Isomorphism th}^{m}]$$

Claim: $\ker\varphi = \langle x^2 - 5\rangle$

$$\langle x^2 - 5\rangle \subseteq \ker\varphi$$

$$f(x) \in \langle x^2 - 5\rangle$$
$$\Rightarrow f(x) = \underline{(x^2 - 5)}\, q(x)$$
$$\varphi(f(x)) = f(\sqrt{5}) = 0\, q(\sqrt{5}) = 0$$

$f(x) \in \ker\varphi$.

$\ker\varphi \subseteq \langle x^2-5 \rangle$

$f(x) \in \ker\varphi \Rightarrow f(\sqrt{5}) = 0$

$(x^2-5) \mid f(x)$ [$\because$ x^2-5 is minimal rel^n of $\sqrt{5}$]

$\Rightarrow f(x) \in \langle x^2-5 \rangle$

$\Rightarrow \ker\varphi \subseteq \langle x^2-5 \rangle$

(a) is proved

(b) Every maximal ideal in $\boxed{\text{commutative ring}}$ with $\boxed{\text{identity}}$ $\Rightarrow$ prime ideal. (*)

Sol^n of part (b)

($\Rightarrow$) $\dfrac{\mathbb{Z}[x]}{\langle x^2-5 \rangle} \cong \mathbb{Z}[\sqrt{5}]$ w/

$= R$

Correspondence Th^m:

Every prime ideal of R $\xleftrightarrow{\text{1-1, onto}}$ Every prime ideals of $\mathbb{Z}[x]$ containing $\langle x^2-5 \rangle$

infact we can show

any prime ideal of R is of the form: $\dfrac{I}{\langle x^2-5 \rangle}$

where I is prime ideal of $\mathbb{Z}[x]$ & $\langle x^2-5 \rangle \subseteq I$.

Let $\tilde{I}$ be that prime ideal

$\Rightarrow 6 \in \tilde{I} \trianglelefteq R \cong \dfrac{\mathbb{Z}[x]}{\langle x^2-5\rangle}$ $\qquad \tilde{\varphi} : \dfrac{\mathbb{Z}[x]}{\langle x^2-5\rangle} \to \mathbb{Z}[\sqrt{5}]$

$\leftarrow 6$

Now, $6 \in \tilde{I} \cong \dfrac{I}{\langle x^2-5\rangle}$

$6 + \langle x^2-5\rangle \in \dfrac{I}{\langle x^2-5\rangle}$ $\Rightarrow 6 \in I$

$\Rightarrow \langle x^2-5\rangle \subseteq I$

$\Rightarrow \langle 6, x^2-5\rangle \subseteq I$

$2 \cdot 3 \in I \Rightarrow$ either $2 \in I$ or $3 \in I$.

$I \overset{?}{\subseteq} \langle 2, x^2-5\rangle \subseteq I$

or $I \overset{?}{\subseteq} \langle 3, x^2-5\rangle \subseteq I$.

Claim: either $I = \langle 2, x^2-5\rangle$

or $I = \langle 3, x^2-5\rangle$

$I = R$.

$$\frac{\mathbb{Z}[x]}{\langle 2, x^2-5\rangle} \cong \frac{\;\frac{\mathbb{Z}[x]}{\langle 2\rangle}\;}{\;\frac{\langle 2, x^2-5\rangle}{\langle 2\rangle}\;} \cong \frac{\mathbb{Z}_2[x]}{\langle x^2-5\rangle}$$

$\cong \mathbb{Z}_2[\sqrt{5}]$

$$\frac{\mathbb{Z}[x]}{\langle 3, x^2-5\rangle} \cong \mathbb{Z}_3[\sqrt{5}] \text{ field.}$$